

GROUP WORK PROJECT # 1
MANAGEMENT
GROUP NUMBER: 9284

MScFE 652: PORTFOLIO

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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).

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Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed.

Note: You may be required to provide proof of your outreach to non-contributing members upon request.

N/A

Step 1

Fama French Factors: 3-Factor Model

The Fama french 3-factor model expands over CAPM asset pricing model by adding size risk and value risk to the market factors i.e. the main factors driving expected returns by investors are sensitivity to market, size and value stocks as measured by book to market ratio.

By including these factors, the model explains the outperforming tendency of assets like stocks and make it a better tool for managers. Any additional unexplained return may be attributed to unsystematic risk.

Also, it has been observed that value and small cap stocks outperform the large cap and growth stocks.

$$R_i - R_f = \alpha_i + \beta_1(R_m - R_f) + \beta_2SMB + \beta_3HML + \epsilon_i$$

where:

- R_i =total return of a stock or portfolio i at time t
- R_f =risk free rate of return at time t
- R_m =total market portfolio return at time t
- $R_i - R_f$ =expected excess return
- $R_m - R_f$ =excess return on the market portfolio (index)
- SMB =size premium (small minus big)
- HML =value premium (high minus low)
- $\beta_1, \beta_2, \beta_3$ =factor coefficients

1.1 Market Risk Premium (Mkt-Rf):

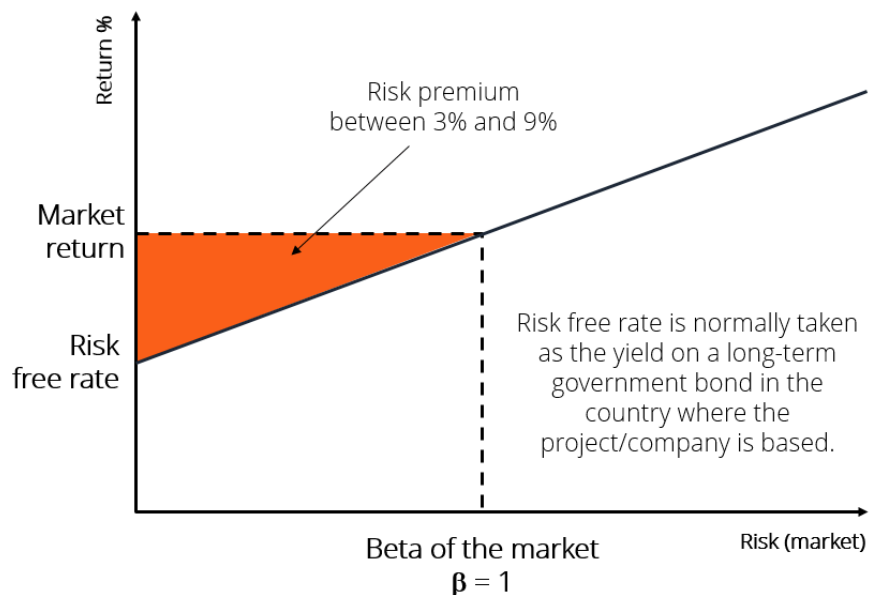
Definition: Market Risk Premium ($R_m - R_f$) is the premium the investors demand for taking the risk and initiative of investment in the market, It is the excess return of market portfolio over the risk free rate.

A stock that is more volatile than the market is considered as riskier and this premium addresses that risk. Investors expects to be reward for taking risk. Some investors would want higher return than a safer or guaranteed return asset like US treasury bills or bonds, and for that they would not mind taking risk for their money.

As per CAPM, an asset's expected return equals the risk-free rate plus compensation for systematic risk (market risk), as measured by beta.

$$E[R_i] = R_f + \beta_i(E[R_m] - R_f)$$

Capital Asset Pricing Model



Courtesy: investopedia

1.2 Size:(SMB: Small minus Big)

Definition: The SMB factor measures the return difference between portfolios of small cap and large cap stocks. It reflects the size effect where small cap companies give higher return than large cap companies.

Small companies are younger, riskier, smaller and less established than big firms. Investors would look for higher return to invest in these companies for taking risk. Historically, Small companies have outperformed big companies especially during times of economic recoveries.

A SMB Bets (Beta₂):above zero suggests that the return in investment portfolio is weighted towards small cap stocks

1.3 Value (HML: High minus Low)

Definition: The HML factor measures the return spread between high book-to-market (value) stocks and low book to market (growth) stocks. It reflects the effect when value stocks outperform the growth stocks.

Companies with high book value relative to their market prices (value stocks) are often struggling or out of favor industries. However they can rebound and deliver good results with

strong returns and come back in business. Investors who invest in value stocks are expecting future recovery. Whereas, stocks with low book to market ratio (growth stocks) are already priced for perfection.

A HML beta (β_{HML}) above zero suggests that the portfolio being measured is weighted towards a high book to market stocks

1.4 Robust (RMW:Robust minus Weak)

This factor compares the robustness of the companies. It is a comparison of the return of companies with high and solid performance to the one with the weak or low profitability operations. It captures how the profitability activities contribute to asset pricing. Firms with robust earnings have a tendency to show a better resilience during downturns, making this factor economically intuitive and practically relevant.

1.5 Investment (CMA:Conservative minus Aggressive)

This factor creates a measure from comparing the firms with conservative investment approaches to the ones with aggressive approaches. The companies with conservative investment generally experience less risks and shocks but their expansion is very limited. Aggressive firms aim for rapid expansion but may face higher risk and instability. This factor helps assess how capital deployment affects expected returns.

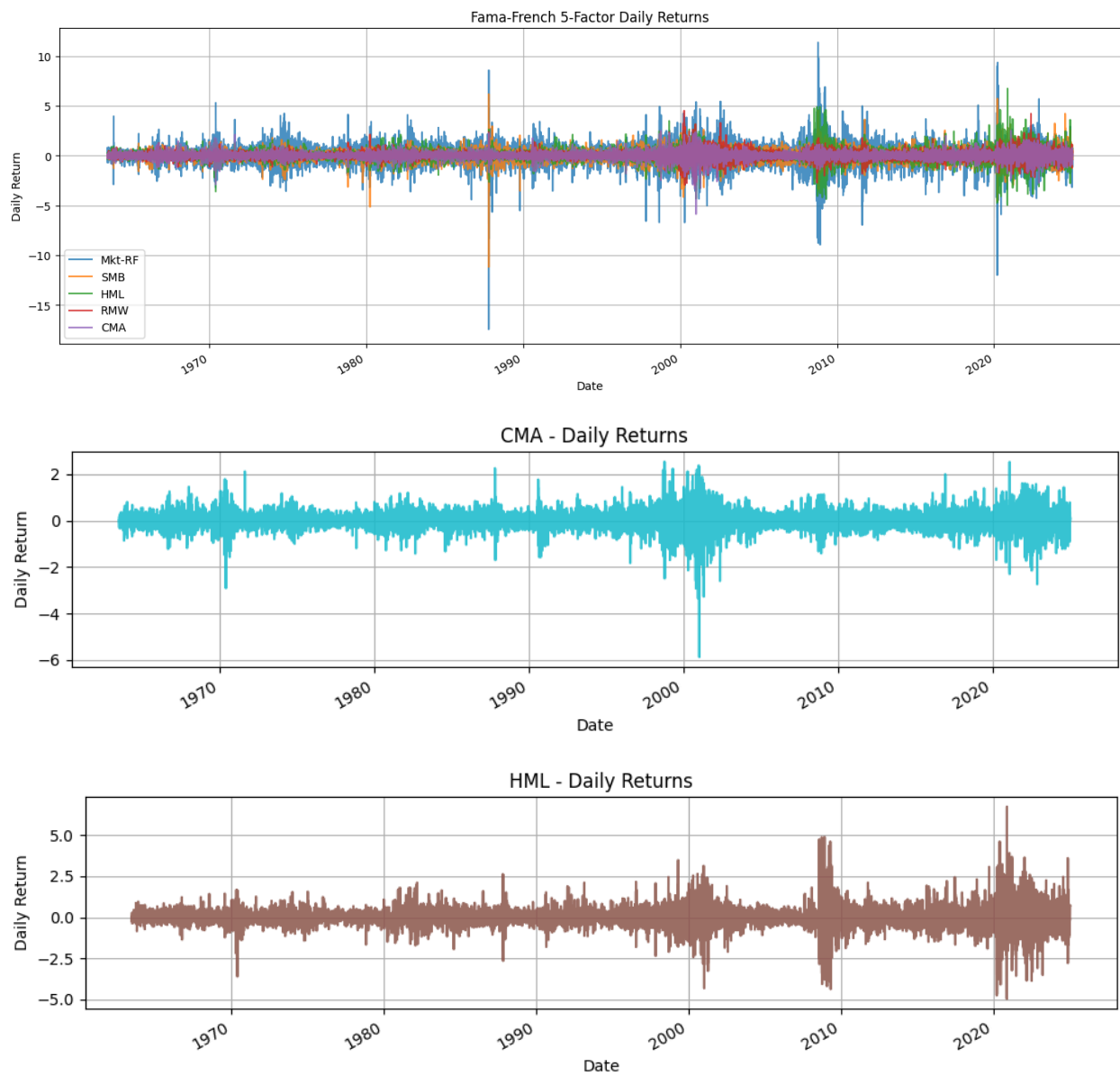
There are views on whether these outperformance tendency is due to market efficiency or market inefficiency. For market efficiency supporters, the outperformance is due to the excess risk that value and small cap face due to their higher cost of capital and greater business risk. For market inefficiency supporters, the outperformance is explained by the market analysts and investors incorrectly pricing the value of these companies, which provides excess return over long run as the value adjusts.

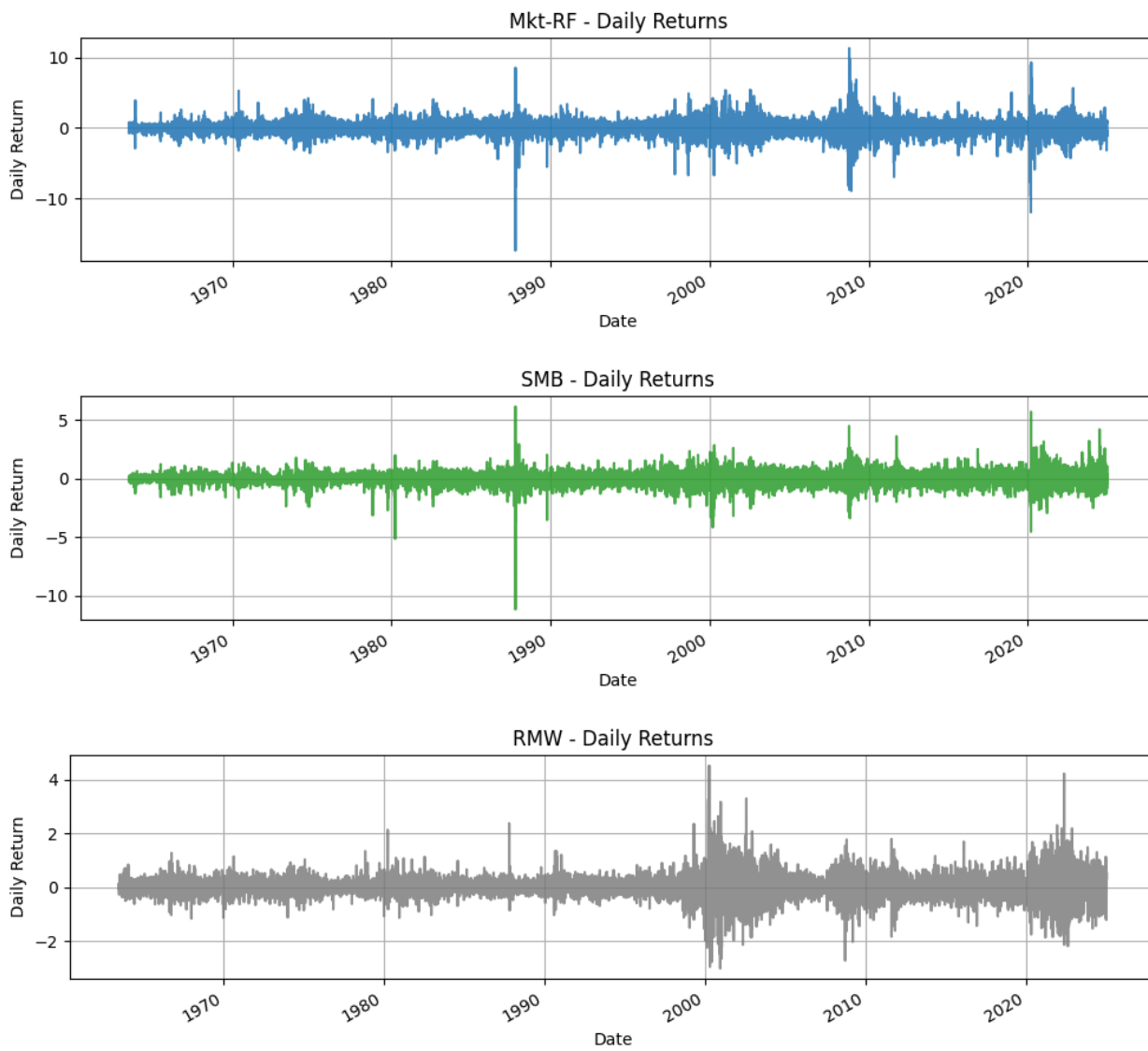
The Fama French model is the result of econometric regression of historical stock prices.

The studies by Fama and French found that the model can explain more than 90% of diversified portfolio returns.

Step 2

2.1 Daily Factor Returns



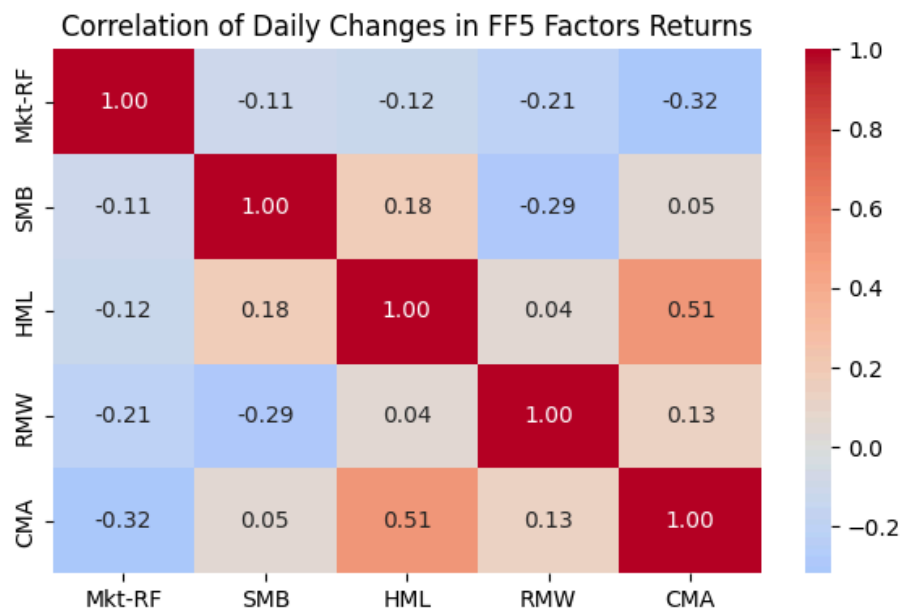


2.2 Correlation and Covariance matrix for changes in factor returns

```
ff5)_delta=ff5.diff().dropna()
```

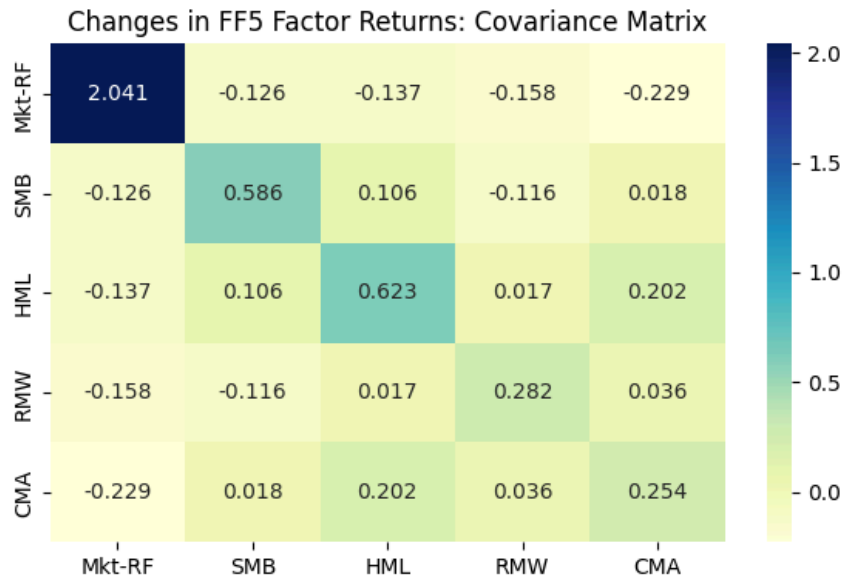
=== Change in FF5 Returns: Correlation Matrix ===

	Mkt-RF	SMB	HML	RMW	CMA
Mkt-RF	1.000	-0.115	-0.121	-0.208	-0.318
SMB	-0.115	1.000	0.176	-0.285	0.048
HML	-0.121	0.176	1.000	0.041	0.509
RMW	-0.208	-0.285	0.041	1.000	0.134
CMA	-0.318	0.048	0.509	0.134	1.000



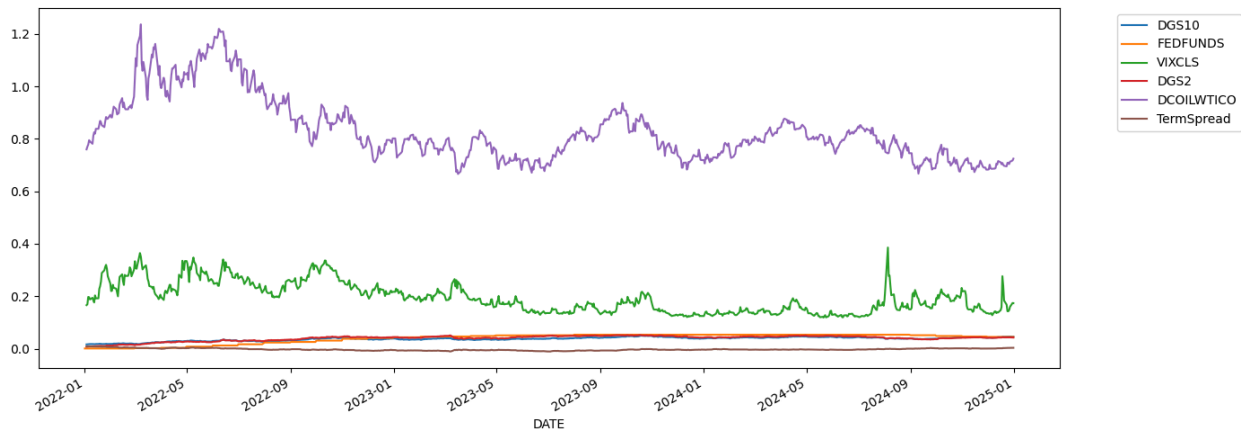
=== Changes in FF5 Returns: Covariance Matrix ===

	Mkt-RF	SMB	HML	RMW	CMA
Mkt-RF	2.041	-0.126	-0.137	-0.158	-0.229
SMB	-0.126	0.586	0.106	-0.116	0.018
HML	-0.137	0.106	0.623	0.017	0.202
RMW	-0.158	-0.116	0.017	0.282	0.036
CMA	-0.229	0.018	0.202	0.036	0.254

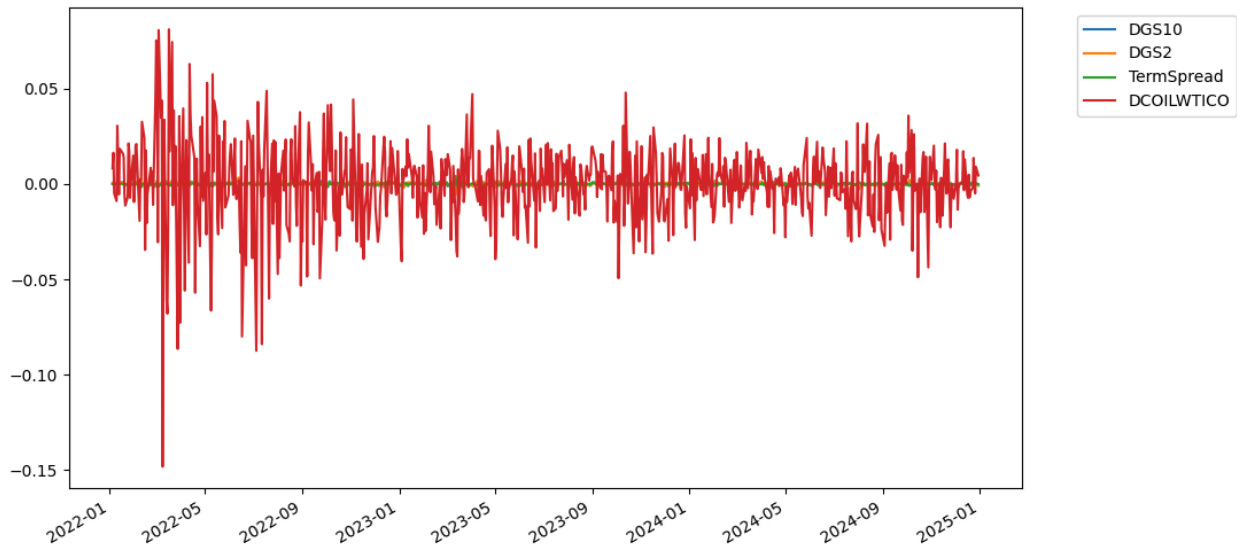


2.3 Download Economic Data of our choice

We downloaded data for DGS10, DGS2, DCOILWTICO and also downloaded FEDFUNDS, VIX as 10-yr yield, 2-yr yield, Crude Oil, EFR, VIX. We calculated term spread as DGS10-DGS2



We took the difference and changed the column names as DGS10_chg, DGS2_chg, DCOILWTICO_chg, TermSpread_Chg to make them stationary. The other macroeconomic factors are already stationary.



Step 3

The `regression()` function we used is a generic function for fitting and comparing a family of factor-based return models (FF3 or FF5, with optional macro data) using any of five regression methods—ordinary least squares, robust Huber, Ridge, Lasso, or ElasticNet— via a single, consistent interface.

Depending on the `train_test_division` flag (80% split), it either performs a one-off “simple” split (fit once on the first part of the data and test on the remainder) or a “rolling” division that refits on a fixed-length window at each step. In the rolling case, it collects the time-series of estimated factor loadings and applies a causal Savitzky–Golay filter to smooth out estimation noise, then uses the final smoothed loadings for out-of-sample prediction.

By z-scoring inputs, automatically adding any remaining DataFrame columns when `macro=True`, and exposing hyperparameters for regularization strength, smoothing window, and model choice, the function makes it easy to do the benchmarking on how different estimation techniques and data sets affect the factor exposures and forecast performance.

We calculated asset Return for TESLA for a 3 year period.

Then we calculated our **Target variable excess return = Asset Return-Risk free Rate** using the returns of the TESLA. We did this analysis for some other assets including the S&P500 but didn't included in the report to don't inflate it size.

We got the risk free rate from the factors data

Features for FF3 Regression: 'Mkt-RF', 'SMB', 'HML' from factors data and 'DGS10_chg', 'RF', 'DGS2_chg', 'DCOILWTICO_chg', 'TermSpread_chg' from macro economic data. Note: we are aware that including the FR as a regressor is not a good practice but we were trying to see if results will get any better-without succeed.

But in both the case for FF3 and FF5 regressions for Tesla we are getting low r squared value (below 35%). Below are the results for FF3 Regression. We tried with various assets but same result i.e. low r squared value

	rmse	r2
huber_simple_ff3	0.036212	0.274668
ols_simple_ff3	0.036229	0.273955
ridge_simple_ff3	0.036234	0.273760
huber_simple_ff3_macro	0.036266	0.272477
ridge_simple_ff3_macro	0.036411	0.266672

The Betas:

ols_simple_ff3	0.016203	0.003727	-0.007817	0.000387	-0.000523	0.001571	0.001284	-0.002035	NaN	NaN
ols_simple_ff3_macro	0.013401	0.000001	0.000599	0.000121	-0.000725	0.001159	0.000532	-0.001740	-0.005984	-0.008619
ols_rolling_ff3	0.015366	0.006275	-0.003397	-0.003230	-0.000012	0.000563	-0.000559	-0.005598	NaN	NaN
ols_rolling_ff3_macro	0.011106	0.010422	-0.002051	-0.003631	-0.003222	0.001061	0.000110	-0.006999	0.007424	-0.005721
huber_simple_ff3	0.015834	0.004362	-0.006902	0.000860	-0.001321	0.001386	0.001743	-0.001046	NaN	NaN
huber_simple_ff3_macro	0.014538	0.000474	-0.000202	0.000796	-0.000441	0.001311	0.000561	-0.001013	-0.006196	-0.005900

$\beta_{\text{Mkt-RF}}$: Sensitivity to the market.

β_{SMB} : Exposure to size (small vs big).

β_{HML} : Exposure to value (high book-to-market vs low).

If $\beta_{\text{Mkt-RF}} \approx 1 \rightarrow$ behaves like the market.

If $\beta_{\text{SMB}} > 0 \rightarrow$ tilted toward small-cap stocks.

If $\beta_{\text{HML}} < 0 \rightarrow$ tilted toward growth stocks.

We tried to run regression on an equally weighted portfolio for multiple assets. Our aim was to understand how our results on the single assets were so bad. As we will explain later, we understood the reason.

We downloaded the asset data as AAPL MSFT, GOOG, AMZN, TSLA for the same 3 year period as factors data. And calculated portfolio returns as

```
asset_returns.dot(equally_weighted_weights)
```

Then, we calculated excess return as portfolio return-Rf (risk free rate)-the return of the portfolio as a whole.

When we ran the FF3 and FF5 regression on the portfolio excess return, we found much better results though it still needs to be improved. **We came to this conclusion that Fama french factors work good for a portfolio return rather than on single asset return. This point comes from the construction of the theory.**

Below are the results for FF3 Regression on portfolio excess return.

	rmse	r2
huber_simple_ff3_macro	0.010056	0.611003
ridge_simple_ff3_macro	0.010290	0.592667
ols_simple_ff3_macro	0.010294	0.592395
ridge_rolling_ff3_macro	0.010417	0.582537
ridge_rolling_ff3	0.010632	0.565165
huber_simple_ff3	0.010693	0.560141
ols_rolling_ff3_macro	0.010770	0.553784
ridge_simple_ff3	0.010833	0.548573
ols_simple_ff3	0.010835	0.548385
ols_rolling_ff3	0.011143	0.522353

The betas:

	Mkt-RF	SMB	HML	RMW	CMA	DGS10_chg	VIXCLS_chg	DGS2_chg	DCOILWTIC0_chg	TermSpread_chg
ols_simple_ff3	0.014299	-0.001454	-0.004224	NaN	NaN	NaN	NaN	NaN	NaN	NaN
ols_simple_ff3_macro	0.013504	-0.001871	-0.002494	-0.000498	-0.002705	0.000069	0.000134	0.000296	0.000495	-0.000389
ols_rolling_ff3	0.013803	0.001478	-0.004698	NaN	NaN	NaN	NaN	NaN	NaN	NaN
ols_rolling_ff3_macro	0.013601	0.002067	-0.004537	0.002288	-0.001393	-0.000801	0.001211	-0.000575	-0.000806	-0.000207
huber_simple_ff3	0.014113	-0.001174	-0.004519	NaN	NaN	NaN	NaN	NaN	NaN	NaN

Step 4

Features for FF5 Regression: 'Mkt-RF', 'SMB', 'HML', 'RMW', 'CMA', from factors data and 'DGS10_chg','RF', 'DGS2_chg', 'DCOILWTICO_chg', 'TermSpread_chg' from macro economic data

FF5 Regression results for TESLA:

	rmse	r2
huber_simple_ff5_macro	0.036266	0.272477
ridge_simple_ff5_macro	0.036411	0.266672
ols_simple_ff5_macro	0.036411	0.266658
huber_simple_ff5	0.036513	0.262535
ols_simple_ff5	0.036754	0.252774

The Betas:

	Mkt-RF	SMB	HML	RMW	CMA	DGS10_chg	VIXCLS_chg	DGS2_chg	DCOILWTICO_chg	TermSpread_chg
ols_simple_ff5	0.014044	-0.000302	0.001579	-0.006270	-0.008954	NaN	NaN	NaN	NaN	NaN
ols_simple_ff5_macro	0.013401	0.000001	0.000599	-0.005984	-0.008619	0.000121	-0.000725	0.001159	0.000532	-0.001740
ols_rolling_ff5	0.014386	0.010242	-0.003205	0.006804	-0.002673	NaN	NaN	NaN	NaN	NaN
ols_rolling_ff5_macro	0.011106	0.010422	-0.002051	0.007424	-0.005721	-0.003631	-0.003222	0.001061	0.000110	-0.006999
huber_simple_ff5	0.015071	-0.000017	0.000923	-0.006517	-0.006195	NaN	NaN	NaN	NaN	NaN

FF5 Regression results for equally weighted Portfolio return:

	rmse	r2
ols_rolling_ff5	0.009872	0.625085
ridge_rolling_ff5	0.009995	0.615725
huber_simple_ff5	0.010002	0.615200
huber_simple_ff5_macro	0.010056	0.611003
ridge_rolling_ff5_macro	0.010144	0.604173

The Betas:

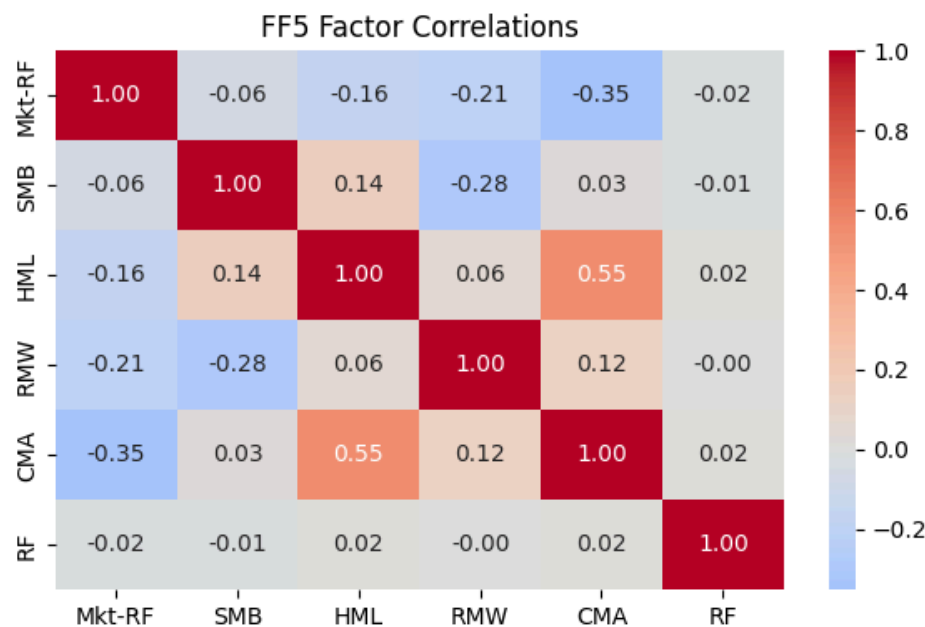
	Mkt-RF	SMB	HML	RMW	CMA	DGS10_chg	VIXCLS_chg	DGS2_chg	DC0ILWTIC0_chg	TermSpread_chg
ols_simple_ff5	0.013521	-0.002003	-0.002105	-0.000636	-0.002800	NaN	NaN	NaN	NaN	NaN
ols_simple_ff5_macro	0.013504	-0.001871	-0.002494	-0.000498	-0.002705	0.000069	0.000134	0.000296	0.000495	-0.000389
ols_rolling_ff5	0.013105	0.002692	-0.004368	0.002294	-0.001599	NaN	NaN	NaN	NaN	NaN
ols_rolling_ff5_macro	0.013601	0.002067	-0.004537	0.002288	-0.001393	-0.000801	0.001211	-0.000575	-0.000806	-0.000207
huber_simple_ff5	0.013183	-0.001544	-0.002426	-0.000354	-0.003045	NaN	NaN	NaN	NaN	NaN

Step 5

5.1 Correlation matrix of Factor Returns

=== FF5 Factor Return Correlation Matrix ===

	Mkt-RF	SMB	HML	RMW	CMA	RF
Mkt-RF	1.000	-0.057	-0.161	-0.211	-0.352	-0.019
SMB	-0.057	1.000	0.141	-0.282	0.028	-0.012
HML	-0.161	0.141	1.000	0.059	0.550	0.017
RMW	-0.211	-0.282	0.059	1.000	0.122	-0.001
CMA	-0.352	0.028	0.550	0.122	1.000	0.016
RF	-0.019	-0.012	0.017	-0.001	0.016	1.000

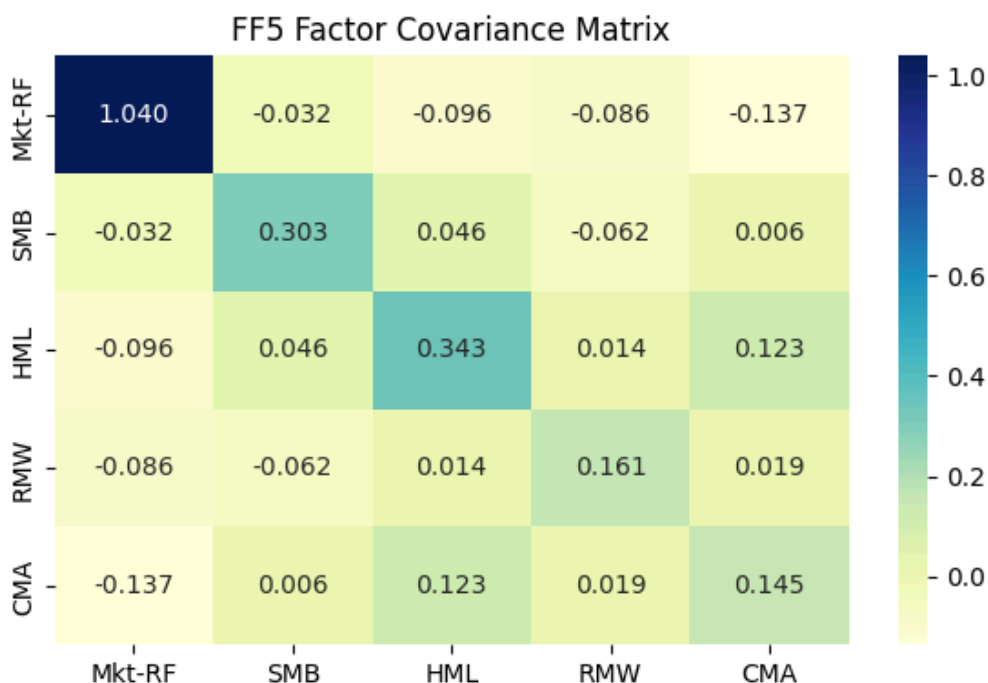


5.2 Covariance matrix of Factor Returns

```

=== FF5 Covariance Matrix ===
      Mkt-RF   SMB   HML   RMW   CMA
Mkt-RF  1.040 -0.032 -0.096 -0.086 -0.137
SMB     -0.032  0.303  0.046 -0.062  0.006
HML     -0.096  0.046  0.343  0.014  0.123
RMW     -0.086 -0.062  0.014  0.161  0.019
CMA     -0.137  0.006  0.123  0.019  0.145

```



Step 6

We can summarize the results of 3-factors and 5-factors model to the Tesla stock in the below two tables

	FF3 Beta	FF5 Beta
Mkt-RF	0.016203	0.013401
SMB	0.003727	0.000001
HML	-0.007817	0.000599
RMW	NaN	-0.005984
CMA	NaN	-0.008619

	FF3	FF5
Adj R ²	0.281871	0.266658
RMSE	0.036031	0.036411

For the equally weighted portfolio the below tables represent the summary of the calculations

	FF3 Beta	FF5 Beta		FF3	FF5
Mkt-RF	0.014433	0.013504	Adj R ²	0.548385	0.592395
SMB	-0.001386	-0.001871	RMSE	0.010835	0.010294
HML	-0.004520	-0.002494			
RMW	NaN	-0.000498			
CMA	NaN	-0.002705			

As we mentioned previously, the R^2 of the factor modeling for a single asset is not acceptable. We understand this as the factors aren't able to explain the return of the stock. For the portfolio the situation is different. Here we used the factors to explain the return of the portfolio as a whole. Here we see that adding more factors to the model improves the explainability of the model and the R^2 increases.

As another approach we could calculate the betas for each asset in the portfolio and then calculate the betas of the portfolio. We did the calculation for this approach too and the above conclusion remains valid.

Step 7

We downloaded the asset data as AAPL MSFT, GOOG, AMZN, TSLA for the same 3 year period as factors data.

Mean Returns:

```
Ticker
AAPL    0.000601
AMZN    0.000640
GOOG    0.000596
MSFT    0.000502
TSLA    0.000799
dtype: float64
```

Covariance Matrix:

```
Ticker    AAPL    AMZN    GOOG    MSFT    TSLA
Ticker
AAPL    0.000292  0.000232  0.000219  0.000203  0.000329
AMZN    0.000232  0.000584  0.000326  0.000289  0.000419
GOOG    0.000219  0.000326  0.000426  0.000249  0.000321
MSFT    0.000203  0.000289  0.000249  0.000302  0.000281
TSLA    0.000329  0.000419  0.000321  0.000281  0.001492
```

We used Markowitz portfolio optimization to find a set of optimal weights for assets.

Markowitz Performance Optimization:

1. Function Portfolio_Performance

Portfolio Expected Return = $\sum w T \mu$

volatility: $\text{squre_root}(\text{weights.T} * \text{cov_matrix} * \text{weights})$

2. The portfolio must achieve the required mean return, (assuming we solve the problem by maximizing the sharpe ratio of the portfolio that achieves the minimum required return).

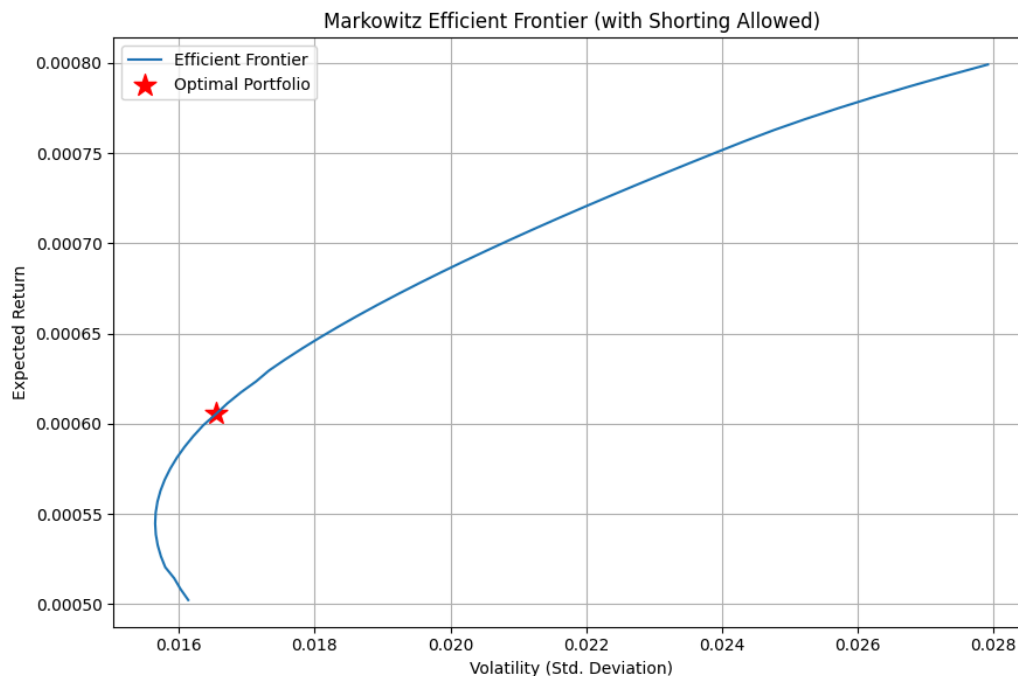
Function Sharpe Ratio = $\text{Portfolio Return} - \text{Risk-Free Rate} / \text{Portfolio Volatility}$

3. We calculated mean return and covariance matrix for the portfolio
4. We used minimize() function to find the portfolio weights that maximize the Sharpe Ratio. But because minimize() minimizes by default, we are minimizing the negative of the Sharpe Ratio.
5. initial_weights: A starting guess for the portfolio weights (e.g., equally weighted).
6. Bounds = [-1.0,1.0] # allow short selling
7. args=mean_returns, cov_matrix, arguments passed to negative_sharpe() to calculate return and volatility.
8. A constraint like $\text{sum}(\text{weights}) = 1$ (fully invested portfolio).
9. SLSQP stands for Sequential Least Squares Programming. It's an optimization algorithm that supports: inequality and equality constraints, bounded variables, smooth nonlinear objective functions

We found optimal weights as

=== Optimal Portfolio Weights ===

	Stock	Weight
0	AAPL	0.676218
1	MSFT	0.104587
2	GOOG	0.153145
3	AMZN	0.038872
4	TSLA	0.027178



We calculated portfolio returns as

```
asset_returns.dot(optimal_weights)
```

Then we calculated excess return as portfolio return- R_f (risk free rate).

It is worthwhile to mention that there is another approach to calculate the betas for the portfolio by betas of each asset and then weighing them. We didn't take this way.

When we ran the FF3 and FF5 regression on the portfolio excess return, we found better results than regression on individual assets though it still needs to be improved. We came to this conclusion that Fama french factors work good for a portfolio return rather than on single asset return.

----- Market Return (Mkt-Rf)-----
Mean Return: 0.028
Standard Deviation: 1.02

----- Portfolio Return (Return*weight)-----
Mean Return: 0.001
Standard Deviation: 0.017

FF3 Regression Results:

We found some regressions are giving good results whereas some are giving low scores for R-squared and there are case of overfitting in certain case where there is good score for train data and low score for test data

	rmse	r2
huber_rolling_ff3_macro	0.009369	0.460332
huber_rolling_ff3	0.009632	0.429605
ridge_rolling_ff3_macro	0.009809	0.408478
ridge_rolling_ff3	0.009889	0.398839
ols_rolling_ff3	0.009948	0.391644

The Betas:

	Mkt-RF	SMB	HML	RMW	CMA	DGS10_chg	VIXCLS_chg	DGS2_chg	DCOILWTIC0_chg	TermSpread_chg
ols_simple_ff3	0.013205	-0.001976	-0.002817	NaN	NaN	NaN	NaN	NaN	NaN	NaN
ols_simple_ff3_macro	0.014257	-0.001082	-0.004750	0.001428	0.001816	0.000198	0.000794	0.000337	0.000313	-0.000269
ols_rolling_ff3	0.011334	0.002146	-0.005603	NaN	NaN	NaN	NaN	NaN	NaN	NaN
ols_rolling_ff3_macro	0.011241	0.001244	-0.007393	0.003626	0.002724	-0.001601	-0.000983	-0.002401	-0.000729	0.001651
huber_simple_ff3	0.012973	-0.001755	-0.003235	NaN	NaN	NaN	NaN	NaN	NaN	NaN

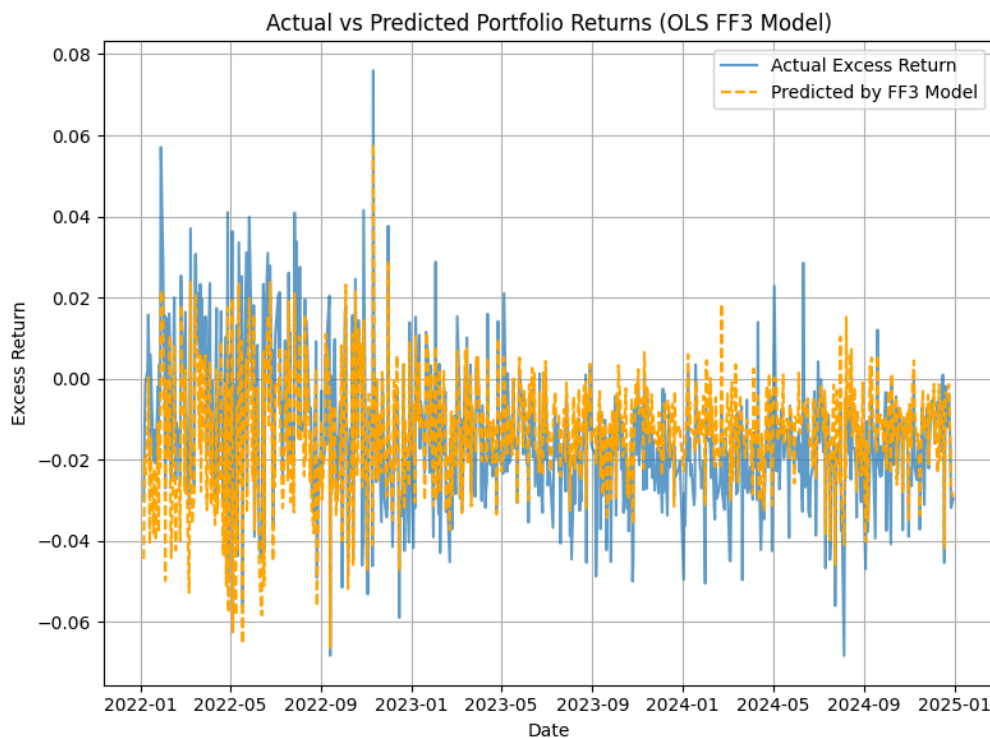
In one case:

OLS Regression Results

Dep. Variable:	Excess Portfolio Return	R-squared:	0.648
Model:	OLS	Adj. R-squared:	0.644
Method:	Least Squares	F-statistic:	155.6
Date:	Thu, 01 May 2025	Prob (F-statistic):	1.02e-129
Time:	08:16:44	Log-Likelihood:	1848.2
No. Observations:	600	AIC:	-3680.
Df Residuals:	592	BIC:	-3645.
Df Model:	7		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-0.0138	0.000	-29.970	0.000	-0.015	-0.013
Mkt-RF	0.0118	0.001	17.953	0.000	0.011	0.013
SMB	-0.0027	0.001	-3.812	0.000	-0.004	-0.001
HML	-0.0031	0.001	-5.448	0.000	-0.004	-0.002
DGS10_chg	0.0933	0.501	0.186	0.852	-0.890	1.076
VIXCLS_chg	0.0494	0.055	0.906	0.365	-0.058	0.156
DGS2_chg	0.4121	0.393	1.048	0.295	-0.360	1.184
DCOILWTICO_chg	0.0059	0.021	0.278	0.781	-0.036	0.048
TermSpread_chg	-0.3188	0.626	-0.509	0.611	-1.548	0.911

Omnibus:	4.422	Durbin-Watson:	0.843
Prob(Omnibus):	0.110	Jarque-Bera (JB):	4.469
Skew:	0.209	Prob(JB):	0.107



----Test Data results for FF3-----
Mean Squared Error (MSE): 0.0001
R-squared (R^2): 0.3546

FF5 Regression Results:

	rmse	r2
ridge_rolling_ff5	0.009044	0.497169
huber_rolling_ff5	0.009068	0.494465
ols_rolling_ff5	0.009284	0.470169
huber_rolling_ff5_macro	0.009323	0.465664
ridge_rolling_ff5_macro	0.009741	0.416634

The Betas:

	Mkt-RF	SMB	HML	RMW	CMA	DGS10_chg	VIXCLS_chg	DGS2_chg	DCOILWTICO_chg	TermSpread_chg
ols_simple_ff5	0.013775	-0.001184	-0.004430	0.001364	0.001647	NaN	NaN	NaN	NaN	NaN
ols_simple_ff5_macro	0.014257	-0.001082	-0.004750	0.001428	0.001816	0.000198	0.000794	0.000337	0.000313	-0.000269
ols_rolling_ff5	0.012580	0.003217	-0.006658	0.003617	0.001595	NaN	NaN	NaN	NaN	NaN
ols_rolling_ff5_macro	0.011241	0.001244	-0.007393	0.003626	0.002724	-0.001601	-0.000983	-0.002401	-0.000729	0.001651
huber_simple_ff5	0.013280	-0.000845	-0.004442	0.001660	0.000711	NaN	NaN	NaN	NaN	NaN

In one case: we found a good score for train data and low score for test data. This is the case of over-fitting. We are unable to understand why some regressions are working fine whereas others are struggling.

```

=====
                        OLS Regression Results
=====
Dep. Variable:      Excess Portfolio Return      R-squared:                0.655
Model:              OLS                        Adj. R-squared:           0.650
Method:             Least Squares              F-statistic:             124.4
Date:               Thu, 01 May 2025            Prob (F-statistic):      4.42e-130
Time:               08:33:58                   Log-Likelihood:          1854.2
No. Observations:   600                       AIC:                     -3688.
Df Residuals:       590                       BIC:                     -3644.
Df Model:           9
Covariance Type:    nonrobust
=====

```

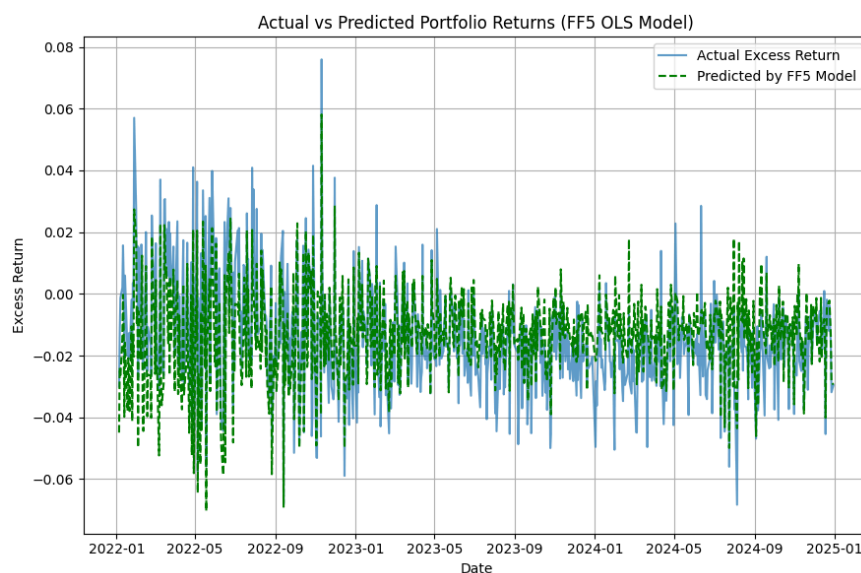
	coef	std err	t	P> t	[0.025	0.975]
const	-0.0138	0.000	-30.243	0.000	-0.015	-0.013
Mkt-RF	0.0123	0.001	17.855	0.000	0.011	0.014
SMB	-0.0015	0.001	-1.855	0.064	-0.003	8.87e-05
HML	-0.0051	0.001	-6.151	0.000	-0.007	-0.003
RMW	0.0022	0.001	2.547	0.011	0.001	0.004
CMA	0.0030	0.001	2.453	0.014	0.001	0.005
DGS10_chg	0.1428	0.497	0.288	0.774	-0.833	1.118
VIXCLS_chg	0.0503	0.055	0.917	0.360	-0.057	0.158
DGS2_chg	0.5566	0.393	1.417	0.157	-0.215	1.328
DCOILWTICO_chg	0.0146	0.022	0.677	0.499	-0.028	0.057
TermSpread_chg	-0.4138	0.622	-0.665	0.506	-1.635	0.808

```

=====
Omnibus:              3.924      Durbin-Watson:              0.840

```

----Test Data results for FF5-----
Mean Squared Error (MSE): 0.0001
R-squared (R^2): 0.3035



Step 8

We know:

$\beta_{\text{Mkt-RF}}$: Sensitivity to the market.

β_{SMB} : Exposure to size (small vs big).

β_{HML} : Exposure to value (high book-to-market vs low).

If $\beta_{\text{Mkt-RF}} \approx 1 \rightarrow$ behaves like the market.

If $\beta_{\text{SMB}} > 0 \rightarrow$ tilted toward small-cap stocks.

If $\beta_{\text{HML}} < 0 \rightarrow$ tilted toward growth stocks.

The variance is not fully explained by the included features in some regressions. Probably, we need to include more relevant features to the dataset for a better R square value like above 75%. We saw that for a single stock the factor modeling doesn't deliver a good explanation but at the portfolio level the ability of the model improves. This is a known fact that comes from the construction of the factors in this approach.

As an interesting point for us, the results of the models didn't get better after adding some macroeconomic data. It shows that designing a factor is not a trivial task.

There are other factor models with more elaborated factors in the literature that can be used to increase the explainability of modeling [6, 7].

Though we have used maximize share ratio as the tool (markowitz optimization) , we can also use minimize variance or risk (Critical Line Algorithm (CLA)) as a tool and plot a full efficient frontier. The PyPortfolioOpt package supports both the cases [8].

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